GUJARAT TECHNOLOGICAL UNIVERSITY

Unit 1: Matrices

Course Code: DI02000011 | Course: Applied Mathematics

Topic 1.1: Concept of a Matrix

Definition: A **matrix** is a mathematical tool that arranges numbers or data into fixed rows and columns enclosed in brackets. It provides a clean way to store and track multiple values at once.

Key Features:

- **Notation:** Usually named with capital letters (like A or B) and written inside square brackets [ ] or round brackets ( ).
- **Dimensions (m × n):** A matrix with m rows and n columns has a size or order of m × n.
- **Address System:** Every number has a specific address based on its row and column position (e.g., a_{12} means row 1, column 2).

Major Uses:

- **Data Organization:** Storing large lists of related data compactly.
- **Programming:** Representing 2D arrays in software code.
- **Physics Modeling:** Tracking forces acting on a structure in multiple directions.

Real-Life Application: Think of a **bus time table**. The rows represent different bus routes, and the columns represent different stops. The times inside the grid are the elements.

Topic 1.2: Types of Matrices

Definition: Matrices come in various shapes and patterns based on how their numbers are arranged. Recognizing these types makes performing mathematical operations much easier.

Key Features:

- **Row/Column Matrix:** A row matrix has only one horizontal row; a column matrix has only one vertical column.
- **Square Matrix:** A matrix where the number of rows exactly equals the number of columns (e.g., 2 × 2 or 3 × 3).
- **Identity Matrix:** A square matrix where diagonal elements are all 1, and all other elements are 0 (acts like the number '1' in normal math).

Major Uses:

- **Simplifying Operations:** Using identity matrices to find the inverse of data sets.
- **Memory Optimization:** Zero matrices (all elements are 0) help clear out system arrays in coding.
- **Structural Engineering:** Square matrices are used to calculate stress and balance in bridge designs.

Real-Life Application: An **Identity Matrix** works exactly like a **mirror**. In engineering software, if you multiply any image matrix by an identity matrix, the output image remains perfectly unchanged.

Topic 1.3: Addition, Subtraction & Scalar Multiplication

Definition: These are the basic arithmetic operations where you combine two matrices together or scale a single matrix up or down using a regular number (scalar).

Key Features:

- **Matching Sizes:** You can only add or subtract matrices if they have the exact same number of rows and columns.
- **Element-by-Element:** Addition and subtraction are done by pairing numbers that sit in the exact same positions.
- **Uniform Scaling:** In scalar multiplication, every single number inside the matrix gets multiplied by that outside number.

Major Uses:

- **Combining Data:** Merging two monthly sales reports into a single matrix.
- **Adjusting Scale:** Scaling a 3D blueprint up or down uniformly in engineering software.
- **Calculating Differences:** Finding the change or error between an expected measurement and an actual measurement.

Real-Life Application: Imagine tracking the **inventory of two stores** (Store A and Store B) in separate matrices. Adding the two matrices together gives you the total inventory for the entire company.

Topic 1.4: Product of Two Matrices

Definition: Matrix multiplication is a specialized way of multiplying two matrices together where rows of the first matrix interact with columns of the second matrix.

Key Features:

- **The Compatibility Rule:** You can only multiply them if the number of columns in the first matrix equals the number of rows in the second matrix.
- **Row-by-Column Process:** You multiply matching elements along a row of the first matrix by a column of the second matrix, then add the results.
- **Order Matters:** Unlike regular math, changing the multiplication order changes the answer ($A \times B$ is usually not equal to $B \times A$).

Major Uses:

- **Combining Transformations:** Linking multiple movements together, like moving a robotic arm forward and then rotating it.
- **Network Calculations:** Figuring out paths and connections in logistics and transport networks.
- **Complex Cost Estimations:** Calculating total material costs when different items have varying prices.

Real-Life Application: Think of a **shopping order layout**. If matrix one lists the quantity of 3 different components needed for 2 projects, and matrix two lists the prices of those components, multiplying them directly calculates the total cost for each project.

Topic 1.5: Adjoint and Inverse of a Matrix

Definition: The inverse of a matrix acts like its mathematical flip or reciprocal (similar to how $^1/_5$ is the reciprocal of 5). The adjoint is a stepping-stone matrix used to calculate this inverse.

Key Features:

- **Square Matrices Only:** You can only find the adjoint and inverse for square matrices (2 × 2 or 3 × 3).
- **The Identity Rule:** Multiplying a matrix by its inverse results in a clean Identity Matrix ($A \times A^{-1} = I$).
- **Determinant Dependency:** If a matrix has a determinant of 0, it is called a singular matrix and cannot be inverted.

Major Uses:

- **Undoing Actions:** Reversing an operation, like clicking 'Undo' after a bad graphic transformation in CAD software.
- **Decryption:** Decoding scrambled messages or securely encrypted data blocks.
- **System Analysis:** Re-engineering electrical circuit currents from output values back to input values.

Real-Life Application: Think of a **digital security key**. The original matrix scrambles your text to hide it, and the inverse matrix is the unique digital key needed to unscramble and read it.

Topic 1.6: Solution of Simultaneous Linear Equations

Definition: This topic teaches you how to use matrices to find the exact values of two unknown variables (like x and y) that satisfy two equations at the exact same time.

Key Features:

- **Matrix Conversion:** Equations are rewritten into a clean formula format: AX = B (where A is the numbers, X is the variables, and B is the answers).
- **Matrix Inversion Method:** You solve for the variables by moving matrix A to the other side as an inverse ($X = A^{-1}B$).
- **Single Intersection:** The solution represents the exact point where two straight lines cross each other on a graph.

Major Uses:

- **Circuit Analysis:** Finding unknown currents flowing through different branches of an electrical circuit.
- **Structural Engineering:** Calculating forces acting on different joints of a bridge truss.
- **Business Balancing:** Finding the break-even point where costs match revenues.

Real-Life Application: Imagine trying to find the individual prices of an **apple and a banana**. If you know that 2 apples and 3 bananas cost ■40, and 1 apple and 4 bananas cost ■35, this matrix method calculates the exact price of each fruit effortlessly.